MATH-101 Calculus and Analytical Geometry – 3-0 CHs

By

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SCALAR TRIPLE PRODUCT

&

VECTOR TRIPLE PRODUCT

Lec # 4

SCALAR TRIPLE PRODUCT OR BOX PPRODUCT

Let $\vec{a}, \vec{b}$ and $\vec{c}$ be any three vectors, then $\vec{a} \cdot \left(\vec{b} \times \vec{c}\right)$ is called scalar triple product of $\vec{a}, \vec{b}$ and $\vec{c}$ and often is denoted by $[\vec{a}\ \vec{b}\ \ \vec{c}\ ]$

Let $\vec{a} = a_1\hat{\imath} + a_2\hat{\jmath} + a_3\hat{k}$

$\vec{b} = b_1\hat{\imath} + b_2\hat{\jmath} + b_3\hat{k}$

$\vec{c} = c_1\hat{\imath} + c_2\hat{\jmath} + c_3\hat{k}$ then

$\vec{a} \cdot \left(\vec{b} \times \vec{c}\right)$ is defined as

$$\left(\vec{b} \times \vec{c}\right) = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$= (b_2\ c_3 - b_3\ c_2)\ \hat{\imath} - (b_1\ c_3 - b_3\ c_1)\ \hat{\jmath} + (b_1\ c_2 - b_2\ c_1)\hat{k}$$

$$\vec{a} \cdot \left(\vec{b} \times \vec{c}\right) = (\mathrm{a}_1 \hat{\mathrm{i}} + \mathrm{a}_2 \hat{\mathrm{J}} + \mathrm{a}_3 \hat{\mathrm{k}}) \cdot ((b_2\, c_3 - \ b_3\, c_2)\, \hat{\imath} - (b_1\, c_3 - b_3\, c_1)\, \hat{\jmath} + (b_1\, c_2 - b_2\, c_1)\hat{k})$$

$$= (b_2\, c_3 - \ b_3\, c_2)\mathrm{a}_1 - (b_1\, c_3 - b_3\, c_1)\mathrm{a}_2 + (b_1\, c_2 - b_2\, c_1)\, \mathrm{a}_3$$

$$\vec{a} \cdot \left(\vec{b} \times \vec{c}\right) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

PROPERTIES

1. The value of scalar triple product remains unchanged if the cyclic order of the vectors remains unaltered, that is

$$\vec{a}\cdot\left(\vec{b}\times\vec{c}\right)=\vec{b}\cdot\left(\vec{c}\times\vec{a}\right)=\vec{c}\cdot\left(\vec{a}\times\vec{b}\right)$$

Now $$\vec{a}\cdot\left(\vec{b}\times\vec{c}\right)=\begin{vmatrix}a_1 & a_2 & a_3\\ b_1 & b_2 & b_3\\ c_1 & c_2 & c_3\end{vmatrix}$$

Interchanging R_1 and R_2

$$=-\begin{vmatrix}b_1 & b_2 & b_3\\ a_1 & a_2 & a_3\\ c_1 & c_2 & c_3\end{vmatrix}$$

Again interchanging R_2 and R_3

$$=\begin{vmatrix}b_1 & b_2 & b_3\\ c_1 & c_2 & c_3\\ a_1 & a_2 & a_3\end{vmatrix}=\vec{b}\cdot\left(\vec{c}\times\vec{a}\right)$$

Similarly for the third case so $\vec{a}\cdot\left(\vec{b}\times\vec{c}\right)=\vec{b}\cdot\left(\vec{c}\times\vec{a}\right)=\vec{c}\cdot$

2. The position of dot and cross can be interchanged without changing the values of scalar triple product, that is

$$\vec{a} \cdot \left(\vec{b} \times \vec{c}\right) = \left(\vec{a} \times \vec{b}\right) \cdot \vec{c}$$

3. If two or three vectors are either equal or parallel then their scalar triple product is zero

$$\vec{a} \cdot \left(\vec{b} \times \vec{c}\right) = 0$$

Using property (2)

$$\vec{a} \cdot \left(\vec{a} \times \vec{b}\right) = (\vec{a} \times \vec{a}) \cdot \vec{b} = 0 \cdot \vec{b} = 0$$

4. $\hat{\imath} \cdot \left(\hat{\jmath} \times \hat{k}\right) = 1$

Since $\hat{\jmath} \times \hat{k} = \hat{\imath}$

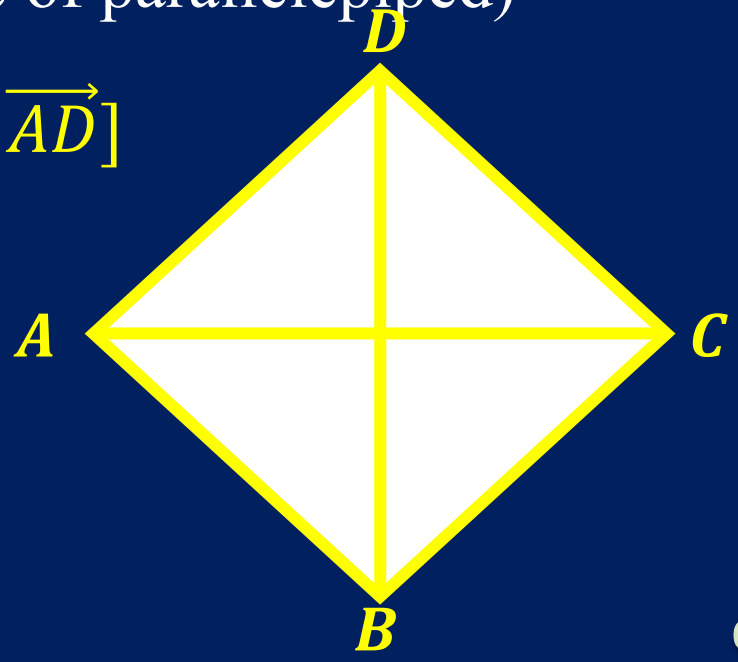
Therefore $\hat{\imath} \cdot \hat{\imath} = 1$

5. The magnitude value of $\vec{a} \cdot \left(\vec{b} \times \vec{c}\right)$ is the volume of parallelepiped (or volume of box)

$$\text{volume of box (parallelepiped)} = \left|\vec{a} \cdot \left(\vec{b} \times \vec{c}\right)\right|$$

6. Volume of tetrahedron $= \frac{1}{6}$ (volume of parallelepiped)

$$= \frac{1}{6}[\overrightarrow{AB} \quad \overrightarrow{AC} \quad \overrightarrow{AD}]$$



EXAMPLE:

Find the volume of the parallelepiped if $\vec{a} = -3\hat{\imath} + 7\hat{\jmath} + 5\hat{k}$, $\vec{b} = -3\hat{\imath} + 7\hat{\jmath} - 3\hat{k}$ and $\vec{c} = 7\hat{\imath} - 5\hat{\jmath} - 3\hat{k}$ are its coterminous edges

SOLUTION: Volume of parallelepiped is $\left|\vec{a} \cdot \left(\vec{b} \times \vec{c}\right)\right|$

$$\vec{a} \cdot \left(\vec{b} \times \vec{c}\right) = \begin{vmatrix} -3 & 7 & 5 \\ -3 & 7 & -3 \\ 7 & -5 & -3 \end{vmatrix}$$

$$= -3(-21 - 15) - 7(9 + 21) + 5(15 - 49) = -272$$

Therefore $\left|\vec{a} \cdot \left(\vec{b} \times \vec{c}\right)\right| = |-272| = 272$ cubic units

Note: Volume is always positive

EXAMPLE:

Find the volume of tetrahedron whose vertices are $A(1,1,1)$, $B(2,1,3)$, $C(3,2,2)$ and $D(3,3,4)$.

SOLUTION:

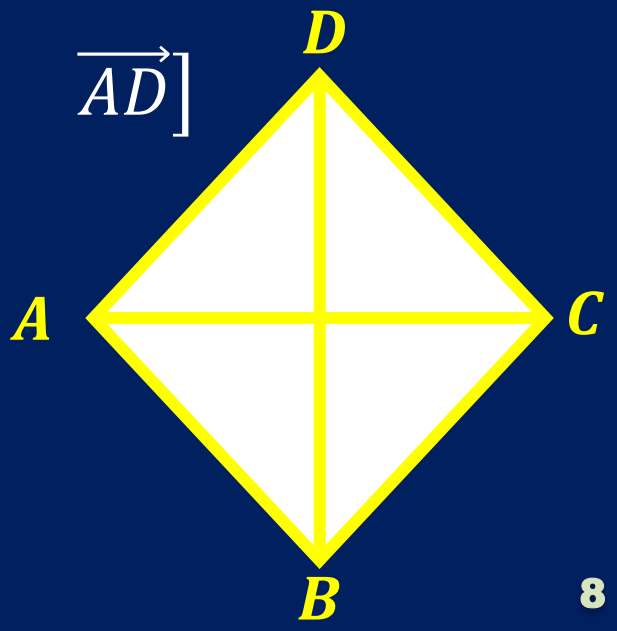
$$\overrightarrow{AB} = (2-1)\hat{\imath} + (1-1)\hat{\jmath} + (3-1)\hat{k} = \hat{\imath} + 2\,\hat{k}$$

$$\overrightarrow{AC} = (3-1)\hat{\imath} + (2-1)\hat{\jmath} + (2-1)\hat{k} = 2\hat{\imath} + \hat{\jmath} + \hat{k}$$

$$\overrightarrow{AD} = (3-1)\hat{\imath} + (3-1)\hat{\jmath} + (4-1)\hat{k} = 2\hat{\imath} + 2\hat{\jmath} + 3\hat{k}$$

Volume of tetrahedron ABCD $= \frac{1}{6}\left[\overrightarrow{AB} \quad \overrightarrow{AC} \quad \overrightarrow{AD}\right]$

$$= \frac{1}{6}\begin{vmatrix} 1 & 0 & 2 \\ 2 & 1 & 1 \\ 2 & 2 & 3 \end{vmatrix} = \frac{5}{6}$$



EXAMPLE: If $\vec{a}, \vec{b}, \vec{c}$ are any three vectors, show that the vectors $\vec{a}-\vec{b},\ \vec{b}-\vec{c},\ \vec{c}-\vec{a}$ are coplanar.

SOLUTION: For coplanar vectors show that their scalar triple product is zero. So we only show that

$$\left(\vec{a}-\vec{b}\right)\cdot\left(\vec{b}-\vec{c}\right)\times(\vec{c}-\vec{a})=0$$

Now

$$\left(\vec{a}-\vec{b}\right)\cdot\left(\vec{b}-\vec{c}\right)\times(\vec{c}-\vec{a})=\left(\vec{a}-\vec{b}\right)\cdot\left\{\vec{b}\times\vec{c}-\vec{b}\times\vec{a}-\vec{c}\times\vec{c}+\vec{c}\times\vec{a}\right\}$$

$$=\left(\vec{a}-\vec{b}\right)\cdot\left\{\vec{b}\times\vec{c}-\vec{b}\times\vec{a}-0+\vec{c}\times\vec{a}\right\}$$

$$=\vec{a}\cdot(\vec{b}\times\vec{c})-\vec{a}\cdot(\vec{b}\times\vec{a})+\vec{a}\cdot(\vec{c}\times\vec{a})-\vec{b}\cdot(\vec{b}\times\vec{c})+\vec{b}\cdot(\vec{b}\times\vec{a})$$

$$-\vec{b}\cdot(\vec{c}\times\vec{a})$$

using property of Cross product

or

using property of scalar triple product (property 2 and 3)

$$= \vec{a} \cdot (\vec{b} \times \vec{c}) + \vec{a} \cdot (\vec{a} \times \vec{b}) - \vec{a} \cdot (\vec{a} \times \vec{c}) - \vec{b} \cdot (\vec{b} \times \vec{c})$$
$$+ \vec{b} \cdot (\vec{b} \times \vec{a}) - \vec{b} \cdot (\vec{c} \times \vec{a})$$

$$= \vec{a} \cdot \vec{b} \times \vec{c} + \vec{a} \times \vec{a} \cdot \vec{b} - \vec{a} \times \vec{a} \cdot \vec{c} - \vec{b} \times \vec{b} \cdot \vec{c}$$
$$+ \vec{b} \times \vec{b} \cdot \vec{a} - \vec{b} \cdot \vec{c} \times \vec{a}$$

$$= \vec{a} \cdot \vec{b} \times \vec{c} - \vec{b} \cdot \vec{c} \times \vec{a}$$

$$= \vec{a} \cdot \vec{b} \times \vec{c} - \vec{a} \cdot \vec{b} \times \vec{c} = 0$$

using property of scalar triple product (property 1)

Q#1 Find the length and direction (when defined) of $\vec{u} \times \vec{v}$ and $\vec{v} \times \vec{u}$.

a) $\vec{u} = 2\hat{i} - 2\hat{j} - \hat{k}$, $\vec{v} = \hat{i} - \hat{k}$

b) $\vec{u} = 2\hat{i} + 3\hat{j}$, $\vec{v} = -\hat{i} + \hat{j}$

c) $\vec{u} = 2\hat{i} - 2\hat{j} + 4\hat{k}$, $\vec{v} = -\hat{i} + \hat{j} - 2\hat{k}$

d) $\vec{u} = \hat{i} + \hat{j} - \hat{k}$, $\vec{v} = 0$

e) $\vec{u} = 2\hat{i}$, $\vec{v} = -3\hat{j}$

f) $\vec{u} = \hat{i} \times \hat{j}$, $\vec{v} = \hat{j} \times \hat{k}$

g) $\vec{u} = -8\hat{i} - 2\hat{j} - 4\hat{k}$, $\vec{v} = 2\hat{i} + 2\hat{j} + \hat{k}$

h) $\vec{u} = \frac{3}{2}\hat{i} - \frac{1}{2}\hat{j} + \hat{k}$, $\vec{v} = \hat{i} + \hat{j} + 2\hat{k}$

Q#2 (a) Find the area of triangle determined by the Points P, Q and R.

(b) Find a unit vector Perpendicular to plane PQR.

1) $P(1,-1,2)$, $Q(2,0,-1)$, $R(0,2,1)$ → Ans (a) $2\sqrt{6}$ (b) $\frac{1}{\sqrt{6}}(2\hat{i}+\hat{j}+\hat{k})$

2) $P(1,1,1)$, $Q(2,1,3)$, $R(3,-1,1)$ → Ans (a) 3 (b) $\frac{1}{3}(2\hat{i}+2\hat{j}-\hat{k})$

3) $P(2,-2,1)$, $Q(3,-1,2)$, $R(3,-1,1)$ → Ans (a) $\sqrt{3}/2$ (b) $\frac{1}{\sqrt{2}}(\hat{i}-\hat{j})$

4) $P(-2,2,0)$, $Q(0,1,-1)$, $R(-1,2,-2)$ → Ans (a) $\sqrt{14}/2$ (b) $\frac{1}{\sqrt{14}}(2\hat{i}+3\hat{j}+\hat{k})$

Q#3 Verify that $(\vec{u}\times\vec{v})\cdot\vec{w} = (\vec{v}\times\vec{w})\cdot\vec{u} = (\vec{w}\times\vec{u})\cdot\vec{v}$ and find the volume of the Parallelepiped (box) determined by $\vec{u}$, $\vec{v}$ and $\vec{w}$.

a) $\vec{u} = 2\hat{i}$, $\vec{v} = 2\hat{j}$, $\vec{w} = 2\hat{k}$ Ans $\boxed{8 = |\vec{u}\times\vec{v}\cdot\vec{w}|}$

b) $\vec{u} = \hat{i} - \hat{j} + \hat{k}$, $\vec{v} = 2\hat{i} + \hat{j} - 2\hat{k}$, $\vec{w} = -\hat{i} + 2\hat{j} - \hat{k}$
Ans $\Rightarrow \boxed{4 = |\vec{u}\times\vec{v}\cdot\vec{w}|}$

c) $\vec{u} = 2\hat{i} + \hat{j}$, $\vec{v} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{w} = \hat{i} + 2\hat{k}$
Ans $\rightarrow \boxed{7 = |\vec{u}\times\vec{v}\cdot\vec{w}|}$

d) $\vec{u} = \hat{i} + \hat{j} - 2\hat{k}$, $\vec{v} = -\hat{i} - \hat{k}$, $\vec{w} = 2\hat{i} + 4\hat{j} - 2\hat{k}$
Ans $\rightarrow \boxed{8 = |\vec{u}\times\vec{v}\cdot\vec{w}|}$

Q#4 Let $\vec{u} = 5\hat{i} - \hat{j} + \hat{k}$, $\vec{v} = \hat{j} - 5\hat{k}$, $\vec{w} = -15\hat{i} + 3\hat{j} - 3\hat{k}$ which vectors if any, are (a) Perpendicular (b) Parallel ? Give reasons for your answers.

Q#5 Let $\vec{u} = \hat{i} + 2\hat{j} - \hat{k}$, $\vec{v} = -\hat{i} + \hat{j} + \hat{k}$,

$\vec{w} = \hat{i} + \hat{k}$, $\vec{r} = -(\pi/2)\hat{i} - \pi\hat{j} + (\pi/2)\hat{k}$. which vectors, if any (a) Perpendicular (b) Parallel? Give reasons for your answers.

Q#6 Find the areas of the Parallelograms whose vertices are given

a) A(1, 0), B(0, 1), C(-1, 0), D(0, -1) $\xrightarrow{\text{Ans}}$ 2

b) A(0, 0), B(7, 3), C(9, 8), D(2, 5) $\xrightarrow{\text{Ans}}$ 29

c) A(-1, 2), B(2, 0), C(7, 1), D(4, 3) $\xrightarrow{\text{Ans}}$ 13

Q#7 Given non-zero vectors $\vec{u}$, $\vec{v}$ and $\vec{w}$, use Dot Product and Cross Product notation, as appropriate, to describe the following

a) The vector Projection of $\vec{u}$ onto $\vec{v}$

b) A vector orthogonal to $\vec{u}$ and $\vec{v}$

c) A vector orthogonal to $\vec{u} \times \vec{v}$ and $\vec{w}$

d) The volume of the Parallelepiped determined by $\vec{u}$, $\vec{v}$ and $\vec{w}$

e) A vector orthogonal to $\vec{u} \times \vec{v}$ and $\vec{u} \times \vec{w}$

f) A vector orthogonal to $\vec{u}+\vec{v}$ and $\vec{u}-\vec{v}$ (Simplify its result) as well)

g) A vector of length $|\vec{u}|$ in the direction of $\vec{v}$.

h) The area of the Parallelogram determined by $\vec{u}$ and $\vec{w}$.

Q#8 Find the areas of the triangles whose vertices are given

a) A(0,0), B(-2,3), C(3,1) $\xrightarrow{\text{Ans}}$ $11/2$

b) A(-1,1), B(3,3), C(2,1) $\xrightarrow{\text{Ans}}$ 2

c) A(-5,3), B(1,-2), C(6,-2) $\xrightarrow{\text{Ans}}$ $25/2$

d) A(-6,0), B(10,-5), C(-2,4) $\xrightarrow{\text{Ans}}$ 42

Q#9 which of the following are always true, and which are not always true? Give reasons for your answers.

a) $\vec{u} \cdot \vec{v} = \vec{v} \cdot \vec{u}$

b) $\vec{u} \times \vec{v} = -(\vec{v} \times \vec{u})$

c) $(-\vec{u}) \times \vec{v} = -(\vec{u} \times \vec{v})$

d) $(c\vec{u}) \cdot \vec{v} = c(\vec{u} \cdot \vec{v})$ (any number c)

e) $c(\vec{u} \times \vec{v}) = (c\vec{u}) \times v = u \times (c\vec{v})$ (any number c).

f) $\vec{u} \cdot \vec{u} = |\vec{u}|^2$

g) $(\vec{u} \times \vec{u}) \cdot \vec{u} = 0$

h) $(\vec{u} \times \vec{v}) \cdot \vec{u} = \vec{v} \cdot (\vec{u} \times \vec{v})$

Q# 10 Let $\vec{u}, \vec{v}$ and $\vec{w}$ be vectors, which of the following make sense, and which do not? Give reasons for your answers

a) $(\vec{u} \times \vec{v}) \cdot \vec{w}$

b) $\vec{u} \times (\vec{v} \cdot \vec{w})$

c) $\vec{u} \times (\vec{v} \times \vec{w})$

d) $\vec{u} \cdot (\vec{v} \cdot \vec{w})$

Q# 11 under what condition the $\vec{a} + \vec{b}$ and $\vec{a} - \vec{b}$ are (i) Perpendicular (ii) Parallel.

Q#12 If $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ and $\vec{b} = 4\hat{j} - 3\hat{k}$, then Show that $\vec{a} \times \vec{b}$ is orthogonal to $\vec{a}$ and $\vec{b}$.

Q#13 Show that $(\vec{a} - \vec{b}) \times (\vec{a} + \vec{b}) = 2(\vec{a} \times \vec{b})$

Q#14 Show that $\vec{a} \times (\vec{b} + \vec{c}) + \vec{b} \times (\vec{c} + \vec{a}) + \vec{c} \times (\vec{a} + \vec{b}) = 0$

Q#15 Find 'x' such that $\vec{a} = \hat{i} + 5\hat{j} + 4\hat{k}$, $\vec{b} = \hat{i} + 2\hat{j} - 3\hat{k}$ $\vec{c} = 3\hat{i} + x\hat{j} + 5\hat{k}$ are Coplanar.

Ans $x = 12$.

VECTOR TRIPLE PRODUCT

If $\vec{A}, \vec{B}$ and $\vec{C}$ are three non -zero vectors, then the vector product of $(\vec{A} \times \vec{B})$ with $\vec{C}$ is called vector triple product and is written as $(\vec{A} \times \vec{B}) \times \vec{C}$ and is defined as

$$(\vec{A} \times \vec{B}) \times \vec{C} = (\vec{C} \cdot \vec{A})\vec{B} - (\vec{C} \cdot \vec{B})\vec{A}$$

Similarly

$$\vec{A} \times (\vec{B} \times \vec{C}) = (\vec{A} \cdot \vec{C})\vec{B} - (\vec{A} \cdot \vec{B})\vec{C}$$

EXAMPLE:

Prove that

$$\vec{a} \times \left(\vec{b} \times \vec{c}\right) + \vec{b} \times (\vec{c} \times \vec{a}) + \vec{c} \times \left(\vec{a} \times \vec{b}\right) = 0$$

Solution:

$$\text{L.H.S} = \vec{a} \times \left(\vec{b} \times \vec{c}\right) + \vec{b} \times (\vec{c} \times \vec{a}) + \vec{c} \times \left(\vec{a} \times \vec{b}\right)$$

$$= (\vec{a} \cdot \vec{c})\vec{b} - \left(\vec{a} \cdot \vec{b}\right)\vec{c} + \left(\vec{b} \cdot \vec{a}\right)\vec{c} - \left(\vec{b} \cdot \vec{c}\right)\vec{a} + \left(\vec{c} \cdot \vec{b}\right)\vec{a}$$

$$-(\vec{c} \cdot \vec{a})\vec{b}$$

$$= 0$$

using property of dot product i.e. Dot product is commutative

EXAMPLE: Prove that

$$\hat{\imath} \times (\vec{a} \times \hat{\imath}) + \hat{\jmath} \times (\vec{a} \times \hat{\jmath}) + \hat{k} \times (\vec{a} \times \hat{k}) = 2\vec{a}$$

Solution : (1st method)

Let $\vec{a} = a_1\hat{\imath} + a_2\hat{\jmath} + a_3\hat{k}$

$$\vec{a} \times \hat{\imath} = (a_1\hat{\imath} + a_2\hat{\jmath} + a_3\hat{k}) \times \hat{\imath} = -a_2\hat{k} + a_3\hat{\jmath}$$

$$\hat{\imath} \times (\vec{a} \times \hat{\imath}) = \hat{\imath} \times (-a_2\hat{k} + a_3\hat{\jmath}) = -a_2(\hat{\imath} \times \hat{k}) + a_3(\hat{\imath} \times \hat{\jmath})$$

$$= a_2\hat{\jmath} + a_3\hat{k} \longrightarrow \textbf{Eq. (1)}$$

Similarly

$$\hat{\jmath} \times (\vec{a} \times \hat{\jmath}) = a_1\hat{\imath} + a_3\hat{k} \longrightarrow \textbf{Eq. (2)}$$

$$\hat{k} \times (\vec{a} \times \hat{k}) = a_1\hat{\imath} + a_2\hat{\jmath} \longrightarrow \textbf{Eq. (3)}$$

Adding Eq. (1), (2) and (3) we have

$$\hat{\imath} \times (\vec{a} \times \hat{\imath}) + \hat{\jmath} \times (\vec{a} \times \hat{\jmath}) + \hat{k} \times (\vec{a} \times \hat{k}) = 2(a_1\hat{\imath} + a_2\hat{\jmath} + a_3\hat{k})$$

$$= 2\vec{a} = R.H.S$$

Solution : (2nd method) Let $\vec{a} = a_1\hat{i} + a_2\hat{J} + a_3\hat{k}$

Using definition of vector triple product

$$L.H.S = \hat{\imath} \times (\vec{a} \times \hat{\imath}) + \hat{J} \times (\vec{a} \times \hat{J}) + \hat{k} \times (\vec{a} \times \hat{k})$$

$$= (\hat{\imath} \cdot \hat{\imath})\vec{a} - (\hat{\imath} \cdot \vec{a})\hat{\imath} + (\hat{J} \cdot \hat{J})\vec{a} - (\hat{J} \cdot \vec{a})\hat{J} + (\hat{k} \cdot \hat{k})\vec{a} - (\hat{k} \cdot \vec{a})\hat{k}$$

$$= \vec{a} - (\hat{\imath} \cdot \vec{a})\hat{\imath} + \vec{a} - (\hat{J} \cdot \vec{a})\hat{J} + \vec{a} - (\hat{k} \cdot \vec{a})\hat{k}$$

$$= 3\vec{a} - (\hat{\imath} \cdot \vec{a})\hat{\imath} - (\hat{J} \cdot \vec{a})\hat{J} - (\hat{k} \cdot \vec{a})\hat{k}$$

$$= 3\vec{a} - a_1\hat{\imath} - a_2\hat{J} - a_3\hat{k} = 3\vec{a} - (a_1\hat{\imath} + a_2\hat{J} + a_3\hat{k})$$

$$= 3\vec{a} - \vec{a} = 2\vec{a}$$

$$\hat{\imath} \cdot \vec{a} = \hat{\imath} \cdot (a_1\hat{\imath} + a_2\hat{J} + a_3\hat{k}) = a_1(\hat{\imath} \cdot \hat{\imath}) = a_1$$
$$\hat{J} \cdot \vec{a} = \hat{J} \cdot (a_1\hat{\imath} + a_2\hat{J} + a_3\hat{k}) = a_2(\hat{J} \cdot \hat{J}) = a_2$$
$$\hat{k} \cdot \vec{a} = \hat{k} \cdot (a_1\hat{\imath} + a_2\hat{J} + a_3\hat{k}) = a_3(\hat{k} \cdot \hat{k}) = a_3$$

EXAMPLE: Prove that

$$(\vec{b}\times\vec{c})\cdot(\vec{c}\times\vec{a})\times(\vec{a}\times\vec{b}) = (\vec{a}\cdot\vec{b}\times\vec{c})^2$$

Solution: Let $\vec{p} = \vec{c}\times\vec{a}$

$$\text{L.H.S} = (\vec{b}\times\vec{c})\cdot\vec{p}\times(\vec{a}\times\vec{b})$$

$$= (\vec{b}\times\vec{c})\cdot\{(\vec{p}\cdot\vec{b})\vec{a} - (\vec{p}\cdot\vec{a})\vec{b}\}$$

$$= (\vec{b}\times\vec{c})\cdot\{(\vec{c}\times\vec{a}\cdot\vec{b})\vec{a} - (\vec{c}\times\vec{a}\cdot\vec{a})\vec{b}\}$$

$$= (\vec{b}\times\vec{c})\cdot(\vec{c}\times\vec{a}\cdot\vec{b})\vec{a}$$

$$= (\vec{b}\times\vec{c}\cdot\vec{a})(\vec{c}\times\vec{a}\cdot\vec{b})$$

$$= (\vec{b}\cdot\vec{c}\times\vec{a})(\vec{c}\cdot\vec{a}\times\vec{b})$$

$$= (\vec{a}\cdot\vec{b}\times\vec{c})(\vec{a}\cdot\vec{b}\times\vec{c})$$

$$= (\vec{a}\cdot\vec{b}\times\vec{c})^2$$

= R.H.S

Now apply the property of scalar triple product that position of dot and cross can be interchange

The value of scalar triple product remains unchanged if the cyclic order of the vectors remains unaltered, that is

$$\vec{a}\cdot(\vec{b}\times\vec{c}) = \vec{b}\cdot(\vec{c}\times\vec{a}) = \vec{c}\cdot$$

ASIDE Let us suppose that $\vec{A} = a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$

$\vec{B} = b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$

$\vec{C} = c_1\hat{i} + c_2\hat{j} + c_3\hat{k}$ then

$$\vec{A} \times \vec{B} = \begin{vmatrix} i & \hat{j} & \hat{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$= (a_2\, b_3 - a_3\, b_2)\,\hat{i} - (a_1\, b_3 - a_3\, b_1)\,\hat{j} + (a_1\, b_2 - a_2\, b_1)\hat{k}$$

$$= (a_2\, b_3 - a_3\, b_2)\,\hat{i} + (b_1\, a_3 - a_1\, b_3)\,\hat{j} + (a_1\, b_2 - a_2\, b_1)\hat{k}$$

$$(\vec{A} \times \vec{B}) \times \vec{C} = \begin{vmatrix} i & \hat{j} & \hat{k} \\ a_2\, b_3 - a_3\, b_2 & b_1\, a_3 - a_1\, b_3 & a_1\, b_2 - a_2\, b_1 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

$$= (b_1 a_3\, c_3 - a_1\, b_3 c_3 - a_1\, b_2 c_2 + a_2\, b_1 c_2)\,\hat{i}$$
$$-(b_3 a_2\, c_3 - a_3\, b_2 c_3 - a_1\, b_2 c_1 + a_2\, b_1 c_1)\,\hat{j}$$
$$+(b_3 a_2\, c_2 - a_3\, b_2 c_2 - a_3\, b_1 c_1 + a_1\, b_3 c_1)\hat{k}$$

$$= (b_1 a_3\, c_3 + a_2\, b_1 c_2)\,\hat{\imath} + (-\,a_1\, b_3 c_3 - a_1\, b_2 c_2)\,\hat{\imath}$$
$$+(a_3\, b_2 c_3 + a_1\, b_2 c_1)\hat{\jmath} + (-b_3 a_2\, c_3 - a_2\, b_1 c_1)\hat{\jmath}$$
$$+ (b_3 a_2\, c_2 + a_1\, b_3 c_1)\hat{k} + (-\,a_3\, b_2 c_2 - a_3\, b_1 c_1)\hat{k}$$

$$= (b_1 a_3\, c_3 + a_2\, b_1 c_2 - a_1\, b_1 c_1 + a_1\, b_1 c_1)\,\hat{\imath} + (-\,a_1\, b_3 c_3 - a_1\, b_2 c_2)\,\hat{\imath}$$
$$+(a_3\, b_2 c_3 + a_1\, b_2 c_1 - a_2\, b_2 c_2 + a_2\, b_2 c_2)\hat{\jmath} + (-b_3 a_2\, c_3 - a_2\, b_1 c_1)\hat{\jmath}$$
$$+ (b_3 a_2\, c_2 + a_1\, b_3 c_1 + a_3\, b_3 c_3 - a_3\, b_3 c_3)\hat{k} + (-\,a_3\, b_2 c_2 - a_3\, b_1 c_1)\hat{k}$$

$$= (b_1 a_3\, c_3 + a_2\, b_1 c_2 + a_1\, b_1 c_1)\,\hat{\imath} + (-\,a_1\, b_3 c_3 - a_1\, b_2 c_2 - a_1\, b_1 c_1)\,\hat{\imath}$$
$$+(a_3\, b_2 c_3 + a_1\, b_2 c_1 + a_2\, b_2 c_2)\hat{\jmath} + (-b_3 a_2\, c_3 - a_2\, b_1 c_1 - a_2\, b_2 c_2)\hat{\jmath}$$
$$+ (b_3 a_2\, c_2 + a_1\, b_3 c_1 + a_3\, b_3 c_3)\hat{k} + (-\,a_3\, b_2 c_2 - a_3\, b_1 c_1 - a_3\, b_3 c_3)\hat{k}$$

$$= (a_1 c_1 + a_2 c_2 + a_3 c_3)\, b_1 \hat{\imath} - (b_1 c_1 + b_2 c_2 + b_3 c_3)\, a_1 \hat{\imath}$$
$$+(a_1 c_1 + a_2 c_2 + a_3 c_3)\, b_2 \hat{\jmath} - (b_1 c_1 + b_2 c_2 + b_3 c_3)\, a_2 \hat{\jmath}$$
$$+(a_1 c_1 + a_2 c_2 + a_3 c_3)\, b_3 \hat{k} - (b_1 c_1 + b_2 c_2 + b_3 c_3)\, a_3 \hat{k}$$

$$= (a_1 c_1 + a_2 c_2 + a_3 c_3)\, (b_1 \hat{\imath} + b_2 \hat{\jmath} + b_3 \hat{k})$$
$$- (b_1 c_1 + b_2 c_2 + b_3 c_3)\, (a_1 \hat{\imath} + a_2 \hat{\jmath} + a_3 \hat{k})$$

$$= (\vec{A} \cdot \vec{C})\vec{B} - (\vec{B} \cdot \vec{C})\vec{A}$$

EXERCISE

Q#1 Show that $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d}) = \vec{a} \cdot (\vec{b} \times \vec{c}) \times \vec{d}$

$$= \begin{vmatrix} \vec{a} \cdot \vec{c} & \vec{c} \cdot \vec{d} \\ \vec{a} \cdot \vec{b} & \vec{b} \cdot \vec{d} \end{vmatrix}$$

Q#2 Show that $\vec{d} \times (\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{c}) = [\vec{a}\ \vec{b}\ \vec{c}](\vec{a} \cdot \vec{d})$

Q#3 Show that $[\vec{a} \times \vec{b}\ \ \vec{b} \times \vec{c}\ \ \vec{c} \times \vec{a}] = [\vec{a}\ \vec{b}\ \vec{c}]^2$

Q#4 Expand $\left[\{ (\vec{a} \times \vec{b}) \times \vec{c} \} \times \vec{d} \right] \cdot \vec{e}$

Q#5 Show that $[\vec{a}+\vec{b} \quad \vec{b}+\vec{c} \quad \vec{c}+\vec{a}] = 2[\vec{a} \quad \vec{b} \quad \vec{c}]$

Q#6 Show that $\vec{a} \times (\vec{b} \times (\vec{c} \times \vec{d})) = (\vec{b} \cdot \vec{d})(\vec{a} \times \vec{c}) - (\vec{b} \cdot \vec{c})(\vec{a} \times \vec{d})$

Q#7 Evaluate $\dfrac{\vec{b} \times \vec{c}}{[\vec{a} \ \vec{b} \ \vec{c}]} \cdot \dfrac{\vec{c} \times \vec{a}}{[\vec{a} \ \vec{b} \ \vec{c}]} \times \dfrac{\vec{a} \times \vec{b}}{[\vec{a} \ \vec{b} \ \vec{c}]}$

Ans $\dfrac{1}{[\vec{a} \ \vec{b} \ \vec{c}]}$

Q#8 Simplify $(\vec{a}+\vec{b}) \cdot (\vec{b}+\vec{c}) \times (\vec{c} * \vec{a})$

Ans $2\vec{a} \cdot (\vec{b} \times \vec{c})$

Q # 9 State whether each expression is meaningful. If not, explain why. If so, state whether it is vector or scalar.

a) $\vec{a} \cdot \vec{b} \times \vec{c}$

(b) $\vec{a} \times (\vec{b} \cdot \vec{c})$

(c) $\vec{a} \times (\vec{b} \times \vec{c})$

(d) $(\vec{a} \cdot \vec{b}) \times \vec{c}$

(e) $(\vec{a} \cdot \vec{b}) \times (\vec{c} \cdot \vec{d})$

(f) $(\vec{a} \times \vec{b}) \cdot (\vec{c} \times \vec{d})$

Q #10 Prove or disprove the formula

$$\vec{u} \times (\vec{u} \times (\vec{u} \times \vec{v})) \cdot \vec{w} = -|\vec{u}|^2 \, \vec{u} \cdot \vec{v} \times \vec{w}$$